Database Management System

Week 7: SQL Query Implementation for E-Commerce Database

Tasks to be Done:

1. **Perform Basic SQL Queries**
   Concept:

SQL SELECT command is used to retrieve data from database tables.

1. **Display all customer details.**

```
SQL> SELECT * FROM CUSTOMER;

                CUSTOMER_ID CUSTOMER_NAME       EMAIL                PHONE          ADDRESS                REGISTRATION_DA
----------------------------- ------------------ ---------------------------- -------------- ---------------------- --------------
                        101 SNEHA              sneha@gmail.com      9876543215     CHENNAI                01-FEB-26
                        102 VIJAY KUMAR        Vijay@gmail.com      9876543216     THIRUNELVELI           05-FEB-26
                        103 ANITHA             anitha@gmail.com                    9876543212     ERODE                  10-FEB-26
                        104 SURESH             suresh@gmail.com                    9876543218     SALEM                  15-FEB-26
                        105 KAMESH             kamesh@gmail.com                    9840371453     KIRABAKKAM             20-FEB-26
```

2. **Display all available products.**

```
SQL> SELECT * FROM PRODUCT;

PRODUCT PRODUCT_NAME
   CATEGOR                PRICE STOCK_QUANTITY
------- ------------------------------
-- ------- ---------- --------------
P01     IDLI
   C01              40             50
P02     BUTTER NAAN
   C02              70             40
P03     CHICKEN FRIED RICE
   C03             140             50
P05     BROWNIE
   C05             200             90
P06     CHICKEN BURGER
   C03             200             18
```

3. **Retrieve product names and prices only.**

```
SQL> SELECT Product_Name, Price FROM Product;

PRODUCT_NAME                                                                    PRICE
--------------------------------------------------------------------------------
-----
IDLI                                                                             40
BUTTER NAAN                                                                      70
CHICKEN FRIED RICE                                                              140
BROWNIE                                                                         200
CHICKEN BURGER                                                                  200
```

4. **Display all orders placed by customers**

```
SQL> SELECT  CUSTOMER_NAME, O.ORDER_ID, O.TOTAL_AMOUNT, O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME      ORDER_ID   TOTAL_AMOUNT ORDER_STATUS
------------------ --------- ------------ ------------------
SNEHA              106              550 Pending
VIJAY KUMAR        107             1250 Delivered
ANITHA             108              480 Preparing
SURESH             109              890 Preparing
KAMESH             110              900 Delivered
```

5. **Retrieve payment details.**

```
PAYMENT ORDER_I PAYMENT_D PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
------- ------- --------- -------------------- ---------- ------------------
PAY101  106     24-SEP-26 UPI                  COMPLETED                3000
PAY102  107     15-MAY-26 CASH                 PENDING                   900
PAY103  108     09-MAR-26 NET BANKING          FAILED                   1000
PAY104  109     10-MAY-26 CARD                 PENDING                  2000
PAY105  110     11-OCT-26 NET BANKING          COMPLETED                1500
```

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT * FROM PRODUCT WHERE PRICE> 5000;

no rows selected

SQL> SELECT * FROM PRODUCT WHERE PRICE> 1500;

no rows selected
```

2. **Display products available in stock**.

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_STATUS = 'Available';

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
I01     P01     S01                50 Available   19-SEP-26
I02     P02     S02                30 Available   14-AUG-26
I03     P03     S03                25 Available   23-JUL-26
I04     P05     S04                35 Available   18-JUN-26
I05     P05     S05                78 Available   05-MAY-26
```

3. **Display completed orders.**

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_STATUS = 'DELIVERED';

no rows selected
```

4. **Find products with rating above 4.**

```
SQL> SELECT P.PRODUCT_NAME, ROUND(AVG(R.RATING),2) AS AVERAGE_RATING
  2  FROM PRODUCT P
  3  JOIN REVIEW R ON P.PRODUCT_ID = R.PRODUCT_ID
  4  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  5  HAVING AVG(R.RATING) > 4
  6  ORDER BY AVERAGE_RATING DESC;

PRODUCT_NAME
--------------------------------------------------------------------
----
AVERAGE_RATING
--------------
CHICKEN FRIED RICE
             5

CHICKEN BURGER
             5

BUTTER NAAN
             5
```

3. Sort Data Using ORDER BY

Tasks:

1. **Display products from lowest to highest price.**

```
SQL> SELECT * FROM PRODUCT ORDER BY PRICE;
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- ----------------
P01
IDLI
C01             40               50

P02
BUTTER NAAN
C02             70               40

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- ----------------

P03
CHICKEN FRIED RICE
C03            140               50

P05
BROWNIE

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- ----------------
C05            200               90

P06
CHICKEN BURGER
C03            200               18
```

2. **Display customers alphabetically.**

```
SQL> SELECT * FROM CUSTOMER ORDER BY CUSTOMER_NAME;

                    CUSTOMER_ID CUSTOMER_NAME      EMAIL          PHONE         ADDRESS              REGISTRATION_DA
----------------------------------- ------------------ ---------------------------- ------------- ---------------------- -------------
                            103 ANITHA           anitha@gmail.com               9876543212    ERODE                10-FEB-26
                            105 KAMESH           kamesh@gmail.com               9840371453    KIRABAKKAM           20-FEB-26
                            101 SNEHA            sneha@gmail.com    9876543215  CHENNAI              01-FEB-26
                            104 SURESH           suresh@gmail.com               9876543218    SALEM                15-FEB-26
                            102 VIJAY KUMAR      Vijay@gmail.com    9876543216  THIRUNELVELI         05-FEB-26
```

3. **Find top expensive products.**

```
SQL> SELECT *
  2  FROM (
  3      SELECT *
  4      FROM PRODUCT
  5      ORDER BY PRICE DESC
  6  )
  7  WHERE ROWNUM <= 3;

PRODUCT PRODUCT_NAME                                                                                                CATEGOR      PRICE STOCK_QUANTITY
------- ------------------------------------------------------------------------------------------------------------------------------------
-- ------- --------- -------------

P05     BROWNIE                                                                                                     C05            200           90
P06     CHICKEN BURGER                                                                                              C03            200           18
P03     CHICKEN FRIED RICE                                                                                          C03            140           50
```

4. **Display latest orders first.**

```
SQL> SELECT * FROM ORDERS ORDER BY ORDER_DATE DESC;

ORDER_ID                                CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------------------------------------- --------- ------------ -------------
110                                            105 10-OCT-26          900 Delivered
106                                            101 23-SEP-26          550 Pending
107                                            102 15-MAY-26         1250 Delivered
109                                            104 10-MAY-26          890 Preparing
108                                            103 09-MAR-26          480 Preparing
```

5. **Retrieve Unique Values Using DISTINCT**

Tasks:

1. Display unique product categories.

```
CATEGORY_NAME
----------------
Beverages
DESSERTS
FAST FOOD
NORTH INDIAN
SOUTH INDIAN
```

2. Find different payment methods used by customers

```
SQL> SELECT DISTINCT PAYMENT_MODE FROM PAYMENT;

PAYMENT_MODE
--------------------
NET BANKING
CASH
CARD
UPI
```

5. **Search Products Based on Conditions**

Tasks: Write SQL queries to:

1. **Find products within a specific price range. (Product Between 130 to 300)**

```
SQL> SELECT * FROM PRODUCT WHERE PRICE BETWEEN 130 AND 300;

PRODUCT PRODUCT_NAME                                                    CATEGOR      PRICE STOCK_QUANTITY
------- ------------------------------------------------------------------------------------------------
-- ------- --------- -------------
P03     CHICKEN FRIED RICE                                              C03           140           50
P05     BROWNIE                                                         C05           200           90
P06     CHICKEN BURGER                                                  C03           200           13
```

2. **Display products belonging to a particular category.**

```
SQL> SELECT * FROM PRODUCT WHERE CATEGORY_ID = 'C03';

PRODUCT PRODUCT_NAME                                                CATEGOR      PRICE STOCK_QUANTITY
------- ------------------------------------------------------------------------------------------------
-- ------- --------- -------------
P03     CHICKEN FRIED RICE                                          C03           140           50
P06     CHICKEN BURGER                                              C03           200           18
```

3. **Find products currently available in inventory**

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_STATUS = 'Available'

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
I01     P01     S01                50 Available   19-SEP-26
I02     P02     S02                30 Available   14-AUG-26
I03     P03     S03                25 Available   23-JUL-26
I04     P05     S04                35 Available   18-JUN-26
I05     P05     S05                78 Available   05-MAY-26
```

4. **Search products using product name.**

```
SQL> SELECT * FROM PRODUCT WHERE PRODUCT_NAME = 'CHICKEN FRIED RICE';

PRODUCT PRODUCT_NAME                                                CATEGOR      PRICE STOCK_QUANTITY
------- ------------------------------------------------------------------------------------------------
-- ------- --------- -------------
P03     CHICKEN FRIED RICE                                          C03           140           50
```

5. **Find low-stock products**

```
SQL> SELECT * FROM INVENTORY WHERE STOCK_QUANTITY < 35;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
I02     P02     S02                 30 Available   14-AUG-26
I03     P03     S03                 25 Available   23-JUL-26
```

6. **Retrieve Customer and Product Information**

Tasks:

1. **Display customer details with their orders.**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,C.EMAIL,C.PHONE,O.ORDER_ID,O.ORDER_DATE, O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOM
ER_ID;

                    CUSTOMER_ID CUSTOMER_NAME       EMAIL           PHONE          ORDER_ID   ORDER_DAT ORDER_STATUS
------------------------------- ------------------- --------------------------- -------------- ---------- --------- --------------------
                            101 SNEHA               sneha@gmail.com    9876543215      106       23-SEP-26 Pending
                            102 VIJAY KUMAR         Vijay@gmail.com    9876543216      107       15-MAY-26 Delivered
                            103 ANITHA              anitha@gmail.com                   9876543212     108       09-MAR-26 Preparing
                            104 SURESH              suresh@gmail.com                   9876543218     109       10-MAY-26 Preparing
                            105 KAMESH              kamesh@gmail.com                   9840371453     110       10-OCT-26 Delivered
```

2. **Display product details with category information.**

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         P.PRICE,
  4         P.STOCK_QUANTITY,
  5         C.CATEGORY_ID,
  6         C.CATEGORY_NAME,
  7         C.DESCRIPTION
  8  FROM PRODUCT P
  9  JOIN CATEGORY C
 10  ON P.CATEGORY_ID = C.CATEGORY_ID
 11  ORDER BY C.CATEGORY_NAME, P.PRODUCT_NAME;
rows will be truncated


PRODUCT PRODUCT_NAME                                                                                                  PRICE STOCK_QUANTITY CATEGOR CATEGOR
------- ------------------------------------------------------------------------------------------------------------------------------------------------
-- ---------- ------------- ------- -------
P05     BROWNIE                                                                                                         200            90 C05     DESSERT
P06     CHICKEN BURGER                                                                                                  200            18 C03     FAST FO
P03     CHICKEN FRIED RICE                                                                                              140            50 C03     FAST FO
P02     BUTTER NAAN                                                                                                      70            40 C02     NORTH I
P01     IDLI                                                                                                             40            50 C01     SOUTH I
```

3. **Find customers who purchased a specific product.**

```
SQL> SELECT DISTINCT C.CUSTOMER_ID, C.CUSTOMER_NAME, C.EMAIL, C.PHONE FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DETAILS OD O
N O.ORDER_ID = OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID WHERE P.PRODUCT_ID = 'P06';

                    CUSTOMER_ID CUSTOMER_NAME       EMAIL           PHONE
------------------------------- ------------------ --------------------------- --------------
                            105 KAMESH             kamesh@gmail.com            9840371453
```

4. **Display products purchased by each customer.**

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      P.PRODUCT_NAME,
  7      OD.QUANTITY,
  8      OD.PRICE,
  9      OD.QUANTITY * OD.PRICE AS LINE_AMOUNT
 10  FROM CUSTOMER C
 11  JOIN ORDERS O
 12      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  JOIN ORDER_DETAILS OD
 14      ON O.ORDER_ID = OD.ORDER_ID
 15  JOIN PRODUCT P
 16      ON OD.PRODUCT_ID = P.PRODUCT_ID
 17  ORDER BY
 18      C.CUSTOMER_ID,
 19      O.ORDER_DATE,
 20      P.PRODUCT_NAME;

                    CUSTOMER_ID CUSTOMER_NAME       ORDER_ID   ORDER_DATE   PRODUCT_NAME           QUANTITY      PRICE LINE_AMOUNT
------------------------------- ------------------ ---------- ------------ ---------------------- ---------- ---------- -----------
                            101 SNEHA              106        23-SEP-26    IDLI                            2        250         500
                            102 VIJAY KUMAR        107        15-MAY-26    BUTTER NAAN                     3        240         720
                            103 ANITHA             108        09-MAR-26    CHICKEN FRIED RICE              5        250        1250
                            104 SURESH             109        10-MAY-26    BROWNIE                                 2        300         600
                            105 KAMESH             110        10-OCT-26    CHICKEN BURGER                  4        400        1600
```

7. **Apply Multiple Filtering Conditions**

   Tasks:

   1. **Find electronics products costing more than ₹10000.**

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, C.CATEGORY_NAME, P.PRICE FROM PRODUCT P JOIN CATEGORY C ON P.CATEGORY_ID = C.CATEGORY_ID WHERE P.PRICE > 100 ORDER
 BY P.PRICE DESC;

PRODUCT PRODUCT_NAME           CATEGORY_NAME                                                                                                    PRICE
------- ---------------------- ------------------------------------------------------------------------------------------------------------------------ ----------
P05     BROWNIE                DESSERTS                                                                                                            200
P06     CHICKEN BURGER         FAST FOOD                                                                                                           200
P03     CHICKEN FRIED RICE     FAST FOOD                                                                                                           140
```

2. **Search products containing the word "BUTTER NAAN".**

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY FROM PRODUCT WHERE UPPER(PRODUCT_NAME) LIKE '%BUTTER NAAN%';

PRODUCT PRODUCT_NAME                PRICE STOCK_QUANTITY
------- ---------------------- --------- -------------
P02     BUTTER NAAN                    70            40
```

3. **Find orders placed within a specific date range.**

```
SQL> SELECT ORDER_ID, CUSTOMER_ID, ORDER_DATE, TOTAL_AMOUNT, ORDER_STATUS FROM ORDERS WHERE ORDER_DATE BETWEEN DATE '2026-03-09' AND DATE '2026-09-23' ORDER
 BY ORDER_DATE;

ORDER_ID                        CUSTOMER_ID ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
--------- ------------------------------------ ---------- ------------ ------------------
108                                      103 09-MAR-26           480 Preparing
109                                      104 10-MAY-26           890 Preparing
107                                      102 15-MAY-26          1250 Delivered
106                                      101 23-SEP-26           550 Pending
```

8. **Generate Basic Business Reports**

Report 1: Product Availability Report

Display:

- Product name
- Price
- Stock quantity
- Availability status

```
SQL> SELECT PRODUCT_NAME, PRICE, STOCK_QUANTITY, CASE WHEN STOCK_QUANTITY >
0 THEN 'Available' ELSE 'OUT OF STOCK' END AS AVAILABIL_STATUS FROM PRODUCT
ORDER BY PRODUCT_NAME;

PRODUCT_NAME                   PRICE STOCK_QUANTITY AVAILABIL_ST
------------------------ ---------- -------------- ------------
BROWNIE                         200             90 Available
BUTTER NAAN                      70             40 Available
CHICKEN BURGER                  200             18 Available
CHICKEN FRIED RICE              140             50 Available
IDLI                             40             50 Available
```

Report 2:

Customer Report Generate:

- **Total number of customers**

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
              5
```

- **New customer registrations**

```
SQL> SELECT CUSTOMER_ID, CUSTOMER_NAME, EMAIL, REGISTRATION_DATE FROM CUSTOMER WHERE REGISTRATION_DATE >= DATE '2026-02-05' AND REGISTRATION_DATE < DATE '20
26-02-15' ORDER BY REGISTRATION_DATE DESC;

                    CUSTOMER_ID CUSTOMER_NAME       EMAIL           REGISTRATION_DA
--------------------------------- ------------------ --------------------------- --------------
                            103 ANITHA             anitha@gmail.com               10-FEB-26
                            102 VIJAY KUMAR        Vijay@gmail.com    05-FEB-26
```

Report 3: Order Report

Display:

- **Total orders**

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS FROM ORDERS;

TOTAL_ORDERS
------------
           5
```

- **Completed orders**

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS FROM ORDERS WHERE ORDER_STATUS = 'Delivered';

COMPLETED_ORDERS
----------------
               2
```

● **Pending orders**

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS FROM ORDERS WHERE ORDER_STATUS IN ('P
reparing', 'pending');

PENDING_ORDERS
--------------
             2
```

● **Cancelled orders**

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS FROM ORDERS WHERE ORDER_STATUS = 'C
ANCELLED';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Find:

● **Highest-priced products**

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE FROM PRODUCT ORDER BY PRICE DESC
;

PRODUCT PRODUCT_NAME                 PRICE
------- ---------------------- ----------
P05     BROWNIE                       200
P06     CHICKEN BURGER                200
P03     CHICKEN FRIED RICE            140
P02     BUTTER NAAN                    70
P01     IDLI                           40
```

- **Most reviewed products**

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, COUNT(R.REVIEW_ID) AS REVIEW_COUNT
 FROM PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID = R.PRODUCT_ID GROUP BY P.PROD
UCT_ID, P.PRODUCT_NAME ORDER BY REVIEW_COUNT DESC;

PRODUCT PRODUCT_NAME              REVIEW_COUNT
------- ------------------------ ------------
P03     CHICKEN FRIED RICE                  1
P01     IDLI                                1
P06     CHICKEN BURGER                      1
P02     BUTTER NAAN                         1
```

- **Available products**

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY FROM PRODUCT WHE
RE STOCK_QUANTITY > 0 ORDER BY STOCK_QUANTITY DESC;

PRODUCT PRODUCT_NAME                PRICE STOCK_QUANTITY
------- ------------------------ ---------- --------------
P05     BROWNIE                        200             90
P01     IDLI                            40             50
P03     CHICKEN FRIED RICE             140             50
P02     BUTTER NAAN                     70             40
P06     CHICKEN BURGER                 200             18
```